

CCW Overview

1. Operating Systems

- **Sources to study:** Operating System Comprehensive Study Materials – Kerala Notes.pdf, CPU Scheduling and semaphore.pdf, CPU-Scheduling-and-Algorithm-(1).pdf, CPU-Scheduling-and-Algorithm-.pdf, Bankers Algo.pdf, Bankers Algo – exercises.pdf, CS401-Bankers-Algorithm-Exercises-Solutions-FINAL.pdf.
- **How to learn:** The CPU scheduling PDFs are highly visual. Focus on practicing the step-by-step Gantt chart calculations for First-Come-First-Served (FCFS), Shortest Job First (SJF), and Round Robin scheduling to master turnaround and waiting times. For deadlocks, work through the Banker's Algorithm exercises by manually calculating the "Need" matrix and finding safe sequences.

2. Database Management Systems (DBMS)

- **Sources to study:** DBMS Comprehensive Study Materials – Kerala Notes.pdf, Normal Forms (1).pdf, Normal Forms.pdf.
- **How to learn:** Start with the comprehensive notes to review the basics of RDBMS, keys (Primary, Foreign, Super, and Candidate), and SQL commands. Then, dive into the Normal Forms documents to memorize the strict criteria for 1NF (atomicity), 2NF (eliminating partial dependency), and 3NF (eliminating transitive dependency).

3. Computer Organization And Architecture (COA)

- **Sources to study:** COA Comprehensive Study Materials – Kerala Notes.pdf, Cache mapping – practice problems.pdf.
- **How to learn:** Use the comprehensive notes to review multiple-choice questions on memory hierarchy, cache mapping techniques, addressing modes, and pipelining hazards.

4. Formal Languages And Automata Theory (FLAT)

- **Sources to study:** FLAT Comprehensive Study Materials – Kerala Notes.pdf.
- **How to learn:** Study the breakdowns covering Deterministic and Non-Deterministic Finite Automata, Context-Free Grammars (specifically Chomsky and Greibach Normal Forms), Push Down Automata (PDA), and Turing Machines. Pay close attention to the closure properties of different language types.

5. Data Structures

- **Sources to study:** WhatsApp Image 2026-05-10 at 9.39.59 AM.jpeg, WhatsApp Image 2026-05-10 at 9.42.19 AM.jpeg, and the general Question Banks (CST308-QP series and

CSE Comprehensive QBANK April 2018).

- **How to learn:** Memorize the Best, Average, and Worst-case time and space complexities for sorting algorithms (like Quicksort, Mergesort, and Bubble Sort) directly from the provided images. Use the comprehensive question banks to practice specific data structure problems, such as calculating the height of binary search trees and evaluating tree traversals.